

CPSC 4300/6300: Assignment 3

Instructions

- Please submit your solution by 11:59PM October 24.
- Submissions should be made through **Canvas**.
- Please complete homework individually.
- Please submit a **single PDF/Word file** including your solutions and the code of your solutions. Feel free to use any third-party libraries.
- **You can only use AI generation tool for Question 1(c).**

1. DBSCAN Algorithm (50 points):

Consider the following Figure 1:

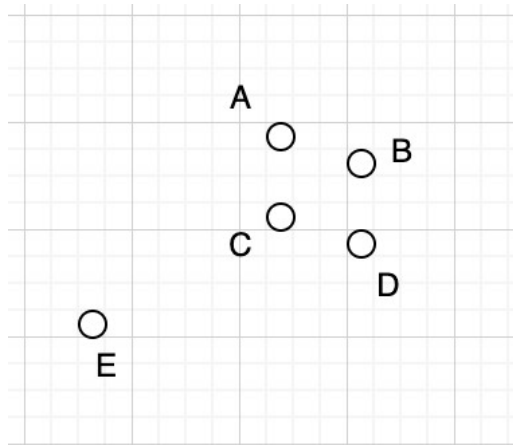


Figure 1: Points

There are 5 points: A, B, C, D, E. The distance matrix between these points is as follows:

	A	B	C	D	E
A	0	2	2	5	6
B	2	0	2	3	7
C	2	2	0	3	6
D	5	3	3	0	7
E	6	7	6	7	0

Table 1: Distance matrix

If we cluster the above points using the DBSCAN algorithm with $\epsilon = 4$ and minimum points = 3,

- (20 points) How many clusters are formed? Draw the outline of your clusters. Explain your reasoning.
- (20 points) With respect to the above figure, state one advantage of DBSCAN over k-means algorithm.
- (10 points) Use an AI generation tool (e.g., ChatGPT, Gemini, or others) to generate a solution to Q(a). Did the AI identify the correct number of clusters? How well did the AI explain its reasoning, is it clear and logically structured? You must cite the AI tool used (e.g., "Generated using ChatGPT (GPT-5) on Oct 15, 2025").

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2. Ensemble Learning (50 points):

Note: This question is based on the `secondary_mushroom_dataset` which can be found in Assignment 3 on Canvas.

- Each record describes a mushroom using physical and biological attributes.
- The target variable class indicates whether the mushroom is **edible (e)** or **poisonous (p)**.
- More descriptions of the features can be found [here](#).

(a) (30 points) Train and evaluate three ensemble models, representing:

- Bagging (Random Forest)
- Boosting (XGBoost)
- Stacking (combine two or more base models with a meta-learner)

Report performance metrics: **Accuracy**, **Precision**, **Recall**, **F1-score**, and **ROC-AUC**.

(b) (20 points) Feature Importance

Using your XGBoost model, identify the top 5 most influential mushroom characteristics that determine edibility or toxicity. **Visualize** feature importance (e.g., bar chart or importance plot). **Interpret** the meaning of these features and discuss whether they make intuitive sense.